AI PM CASE STUDY

Fundpedia

A mutual fund assistant that answers with a fact and a source, or says it does not know.

HDFC mutual fund FAQ assistant

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01 THE PROBLEM

Awareness is high. Confidence is low.

Households that know about mutual funds or ETFs



53%

Households that actually invest in them



6.7%

27%
of Indian adults are financially literate

74%
of fund investors prefer low risk

More than 10 crore SIP accounts are active. Many owners are careful beginners who lack an answer they can trust.

Sources: SEBI Investor Survey 2025, NCFE, AMFI (Aug 2026)

02 WHAT USERS SEARCH

The questions people ask are simple

Question	Answer lives in	Why it hurts
What is the exit load?	Scheme document	A wrong number costs money
What is the lock-in period?	Scheme page	Rules differ by scheme type
What is the NAV today?	Fund page	It changes every business day
Which benchmark does it track?	Factsheet	Needed to judge it fairly
What is the expense ratio?	Scheme page	Small number, big effect
How do I get my statement?	Help pages	Everyone asks, few find it

None of these need advice. Each one needs a fact and a source.

Question types I designed around. Real chat logs will refine this list.

03 USER INTERVIEWS

Two traders told me where they get stuck

Rohan, 29, intraday trader

First SIP last month, saving for a house down payment

"I read a stock chart in seconds. A fund factsheet takes me twenty minutes."

"I never know if the exit load applies to me. I just hope not."

"Do not tell me what to buy. I trade for a living, I do not need a tip."

Built: answer first, source linked, exit load stated plainly, advice refused.

Neha, 33, swing trader

Checking funds for her parents' savings

"I typed which HDFC fund has low charges. I got a wall of text."

"My father holds a fund. I only remember the name starts with HDFC."

"If I cannot see where a number came from, I will not use it."

Learned: users skip the fund name and compare across funds.

Illustrative interviews with two trader users. Replace with notes from real conversations.

04 WHY IT IS HARD

Three risks, three rules

RISK 1

Confident and wrong

Funds have similar names. A mixed up exit load costs real money.

RULE 1

No source, no answer. Every answer carries a citation.

RISK 2

Accidental advice

Telling someone what to buy is regulated advice.

RULE 2

Check the question first. Advice stops before the model sees it.

RISK 3

Private data in chat

People paste PAN or phone numbers without thinking.

RULE 3

Scan every message. Personal data is blocked before it travels.

One bad answer costs more trust than ten good answers earn.

05 HOW IT IS BUILT

Two paths: one for answers, one for data

ANSWER PATH

Browser app
HTML, CSS and JavaScript on Vercel

→

API
FastAPI on Railway runs every check

→

Search
Qdrant Cloud, with BGE embeddings from Cloudflare

→

Answer model
An LLM on Groq writes the reply

DATA PATH

Groww pages
5 HDFC fund pages

→

Scraper
Playwright, run daily by GitHub Actions

→

Chunker
35 chunks of 250 tokens

→

Index
Vectors saved in Qdrant Cloud

Everything runs on free tiers, so the whole stack costs $0 a month.

05 HOW IT IS BUILT

The stack and who provides each part

Part	What I used	Why
Frontend	HTML, CSS, JavaScript on Vercel	No build step, free
Backend	Python and FastAPI on Railway	Small API, free tier
Vector database	Qdrant Cloud (Chroma locally)	Managed cosine search
Embeddings	BGE large, 1024 dims, Cloudflare	Free tier, runs hosted
LLM	Groq, OpenAI compatible, temperature 0	Fast and repeatable
Data fetch	Playwright reads Groww page data	Pages load with scripts
Scheduler	GitHub Actions, daily at 10:00 AM IST	Free runner minutes

Dev mode uses local Chroma and local embeddings, so I can build without paying anything.

05 THE DATA SOURCE

Where the data comes from, and how it stays fresh

SOURCE

5 HDFC fund pages on Groww

groww.in/mutual-funds/hdfc-mid-cap-fund-direct-growth

groww.in/mutual-funds/hdfc-large-cap-fund-direct-growth

groww.in/mutual-funds/hdfc-small-cap-fund-direct-growth

groww.in/mutual-funds/hdfc-gold-etf-fund-of-fund-direct-plan-growth

groww.in/mutual-funds/hdfc-defence-fund-direct-growth

Only these 5 pages are allowed. The bot answers from nothing else.

DAILY REFRESH

Cron job on GitHub Actions

1. Every day at 10:00 AM IST (04:30 UTC), the cron starts the workflow.
2. Playwright opens each page and reads its page data.
3. The text is split into sections and 250 token chunks.
4. Vectors are rebuilt in Qdrant Cloud.
5. Chat pauses until the update finishes.

No one has to remember to update the data. It refreshes itself every day.

Locally, an in-app scheduler also runs at 09:15 AM IST. It is off on Railway, where GitHub Actions does the work.

06 HOW A QUESTION IS HANDLED

Nine steps, and most run before the model

BEFORE THE MODEL

Rate limit
20 requests a minute per user

→

Clean input
Cut to 500 characters, strip injection phrases

→

PII scan
Block PAN, Aadhaar, phone, email, OTP, bank numbers

→

Classify
Rules sort it: fact, advice, small talk or off topic

→

Clarify
No fund named? Show 5 fund buttons

WITH THE MODEL

Rewrite
Fix pronouns and nicknames using the last 10 messages

→

Retrieve
Top 4 passages above 0.35 similarity, for the named fund

→

Answer
Three sentences at most, from the passages only

→

Check
Every link must match a retrieved source, or it is removed

Before step 1, a sync lock pauses chat while the data updates.

07 THE KNOWLEDGE BASE

Small on purpose, so I can check it by hand

5
HDFC schemes

35
chunks of 250 tokens

4
passages per question

0.35
minimum similarity

Schemes covered

- Mid Cap
- Large Cap
- Small Cap
- Gold ETF Fund of Fund
- Defence

Sections saved per fund

- Overview
- Fund details
- Exit load
- Fund managers
- Returns
- About
- Investment objective

All five are Direct Growth plans. Each chunk keeps its scheme, section, source link and scrape time.

08 DECISIONS

Every guardrail trades convenience for trust

Decision	What I gained	What I gave up
Refuse advice questions	Stays factual and safe	Some users get no answer
Say I do not know	No made up answers	Fewer questions answered
Pause chat during sync	No half updated answers	A 2 to 3 minute wait
Start with 5 HDFC schemes	Data I can check by hand	Narrow coverage
Rules to classify questions	Fast and testable	Misses unusual wording
Free tiers everywhere	$0 a month	Slow first request

> On screen: one page with no scrolling, a black and green theme, and a visible sync status.

Every row gives up some convenience. I chose trust each time.

Live demo: hdfc-mutual-fund-faq-groww.vercel.app

09 RESULTS

What the tests show, and what they miss

What passed

Unit tests	78 of 78
Edge case questions	41 of 41
Advice that slipped through	0
Blank responses	0

Not measured yet: full response time

Known weak spots

Groww layout changes can break the scraper.

Holdings and asset mix are not scraped yet.

Only 4 passages, so long multi part questions lose parts.

Chat history resets when Railway redeploys.

Source: my pytest suite and a 41 case edge test document across 6 question categories.

10 WHAT COMES NEXT

Next: understand the question better

1 ANSWER

In the knowledge base

“What is the exit load of HDFC Mid Cap Fund?”

TODAY

Works. Cites the source.

NEXT

Cover long questions that need more than 4 passages.

2 ASK BACK

Unclear question

“Which HDFC fund has low charges?”

TODAY

Only a missing fund name triggers a question.

NEXT

Ask what a person would ask, or compare all 5 funds.

3 DECLINE

Cannot answer

“Should I put my savings in this fund?”

TODAY

Rules catch known advice phrases.

NEXT

A model catches odd wording and Hinglish.

Also next: Hindi support and more fund houses. Yash Vijayvargiya | linkedin.com/in/yv12